

Adaptive DRL-Driven Communication Protocols for Enhanced Data Delivery in Vehicular Ad Hoc Networks

PhD Proposal Submitted to



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1. Introduction

1.1 Introduction to VANET

The Vehicular Ad-Hoc Network (VANET), a particular type of Mobile Ad-Hoc Network (MANET), is a crucial part of the Intelligent Transportation System (ITS) and enables communication between adjacent automobiles and roadside infrastructure[1][2]. In order to create a self-organising network without being aware of or identifying each other beforehand. Cars are used as mobile nodes in this network. By acting as wireless routers or nodes, participating vehicles can establish connections with others that are within close range—typically around 100 to 300 meters in urban settings and approximately 1000 meters on highways [2]. Although nodes may occasionally lose their connection, they can reconnect, which leads to a dynamic internetwork among vehicles [3].

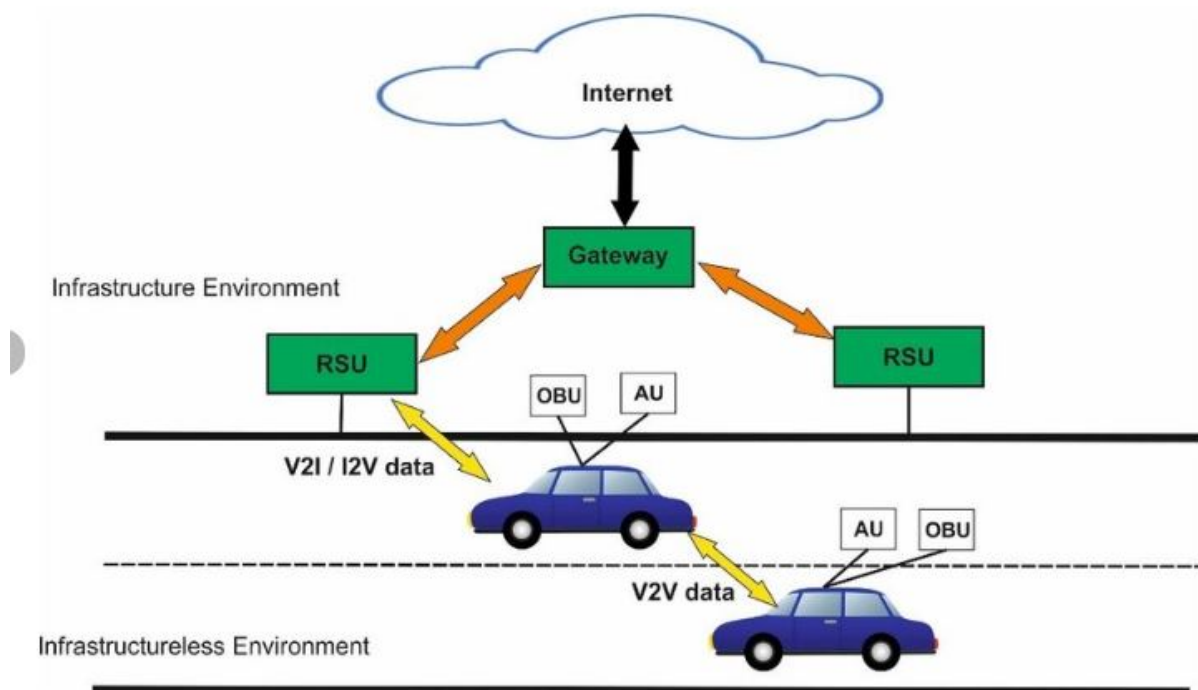


Figure 1.1: Basic VANET Architecture

Figure 1.1 depicts the fundamental structure of a VANET, showcasing how vehicles interact with one another Vehicle to Vehicle (V2V) and with roadside infrastructure Vehicle to Infras-

structure/ Infrastructure to Vehicle(V2I/I2V) through On-Board Units (OBU) and Application Units (AU), facilitating data transmission via Road Side Units (RSUs) and a gateway linked to the internet.

VANET features unique characteristics, including a geography-constrained topology, predictable patterns of movement, varying vehicle density, and changing channel capacity, which distinguish it as a significant research area within MANET [4]. While sustaining Quality of Service(QoS) and allowing infotainment applications, these features raise a number of challenges associated with medium access, physical layer communication, routing, congestion control, fault tolerance, end to end data transmission, the regulation of both safety and non-safety information.[1][3].

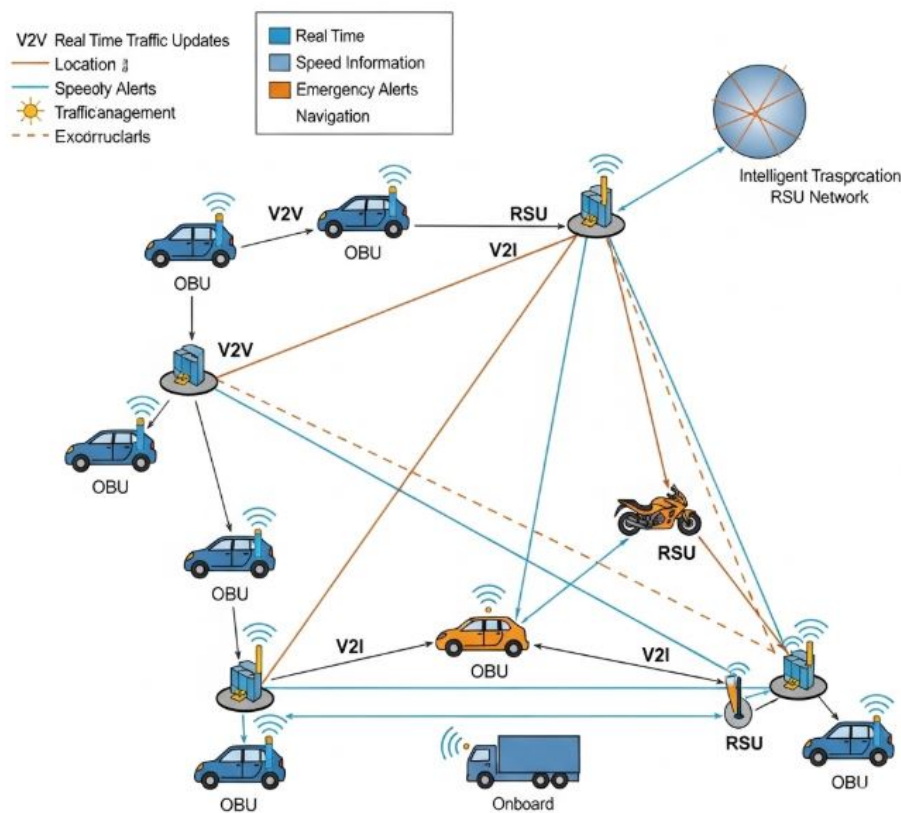


Figure 1.2: VANET for Data Communication

Figure 1.2 illustrates the way vehicles utilize VANET to interact with one another to share real-time traffic information, speed details, emergency notifications, and navigation assistance, all integrated into an intelligent transportation system.

VANET is made up of groups of vehicles, whether in motion or stationary, that are interconnected via a wireless network. Recently, the primary function of VANETs was to enhance safety and provide comfort for drivers in vehicular settings. This perspective is evolving: VANETs are now recognized as a crucial component of an intelligent transportation system, particularly with the rise of autonomous vehicles and the need for Internet access in smart cities. Furthermore, VANETs enable the onboard computers of mostly stationary vehicles,

such as those in an airport parking lot, to act as resources for a mobile computing cloud with minimal reliance on Internet infrastructure.

Messages in VANETs can be broadly classified into priority messages, which require differentiated handling, and duplicate messages, which introduce redundancy and bandwidth wastage. Priority messages, such as collision warnings and emergency braking alerts, demand immediate channel access and minimal delay, whereas non-safety messages, including infotainment or commercial advertisements, can tolerate higher latency. Failure to distinguish between these message types can compromise road safety and degrade overall network performance.

VANETs provide the sharing of a variety of information, from non-safety entertainment data to life-threatening safety alarms. The urgency, relevancy, and temporal sensitivity of these communications vary greatly. However, under congested network conditions, traditional VANET communication protocols frequently handle all messages equally, delaying the delivery of safety-critical information.

Another major challenge in VANET communication is the presence of duplicate messages. Due to broadcast-based dissemination, multi-hop forwarding, and high vehicle density, the same information is often retransmitted multiple times by neighboring vehicles. This redundancy leads to unnecessary bandwidth consumption, increased channel contention, and packet collisions, commonly referred to as the broadcast storm problem.

Information generated and utilized by vehicles has significance that is specific to its time, location, and the parties involved, namely the producer and the consumer. Data produced by vehicles holds local significance, has a restricted geographical range, a defined duration, a limited timeframe, and is of local interest, being pertinent to individuals within a confined area surrounding the vehicle. For instance, the notification that a car is nearing a congested section of a highway is applicable solely to that specific part of the roadway, at that moment, and for nearby vehicles.

VANETs serve as the communication framework for ITS, traffic, and entertainment information in real-time. Nevertheless, High mobility presents issues for VANETs., changing topologies, and Regular link disruptions, which complicate dependable and efficient data transmission. Traditional fixed or rule-oriented communication protocols often find it difficult to adapt to the dynamic VANET environment.

In order to guarantee the timely delivery of vital information while preserving bandwidth efficiency in extremely dynamic VANET systems, efficient message priority categorisation and duplicate message suppression are essential.

1.2 Introduction to Deep Reinforcement Learning (DRL)

By allowing agents (vehicles) to make the best decisions through continuous interaction with their environment, DRL presents a convincing way to support adaptive communication tactics. reference [5]. The goal of this research is to develop DRL-powered adaptive communication

protocols that will improve data transmission, lower latency, and increase reliability in VANETs under various traffic conditions.

DRL is a powerful machine learning technique which combines the complex pattern recognition abilities of Deep Learning (DL) with making decisions power of RL [6]. In the data transmission domain, DRL offers a way to create intelligent and flexible network systems that can improve their performance in unpredictable and uncertain circumstances.

DRL enables intelligent network agents, such as controllers or routers, to make well informed decisions through engaging with the communication surroundings. The agent evaluates the network's condition, accounting for variables like traffic, bandwidth, and signal quality, before taking any action, such as modifying routes or assigning resources. It is rewarded based on the result of each action (e.g., reduced latency or increased throughput) and uses this information to make better decisions going forward.

DRL employs deep neural networks to maximise long-term gains over time to assess the most effective strategy or guideline. For handling dynamic network tasks like routing, congestion control, power management, handovers, anomaly detection, and edge computing, DRL is especially well-suited. It is especially crucial for the management of current and future communication networks, such as 5G and VANET, because of its capacity to learn, adapt, and maximise in complex network contexts.

Figure 1.3 illustrates Vehicle movement, channel quality, and message urgency are just a few of the real-time data that the adaptive DRL agent continuously receives from the VANET environment. It then uses this information to intelligently choose the best communication actions, such as prioritization, routing, and bandwidth control, and implements them in the network. Metrics for performance, including packet delivery ratio, latency, and throughput, are utilized to evaluate the outcomes, and the adaptive DRL agent uses the feedback to learn and improve future communication decisions for more accurate and reliable data transmission.

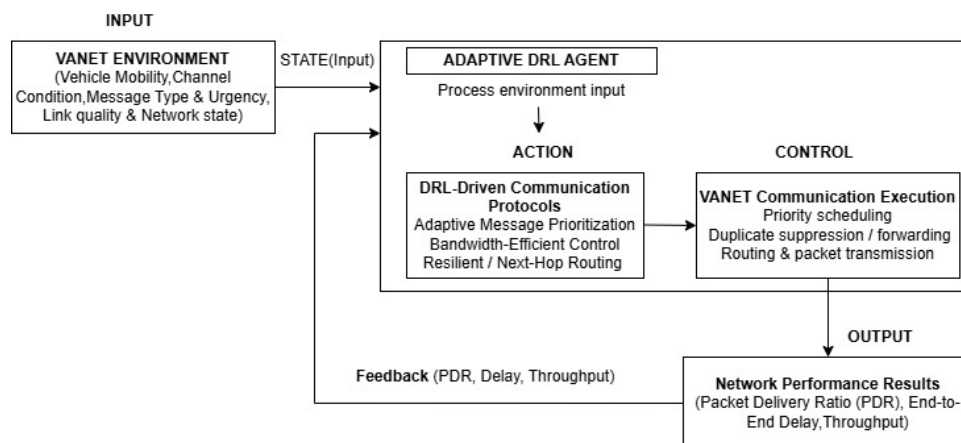


Figure 1.3: Architecture for Adaptive DRL Driven Communication Protocols

1.3 Identified Problems in this Area

VANETs transmit and receive data rapidly while cars are in motion by utilising wireless technology. However, issues like data loss, delays, and connection failures can be brought on by rapidly changing road conditions, fast-moving vehicles, and constrained network resources [7][8]. Traditional routing and communication protocols often struggle to adapt to VANET's dynamic environment, resulting in packet loss, increased delays, and reduced network throughput [9].

Some of the identified problems are:

- **Channel Congestion:** The channel gets clogged as more cars join, seriously impeding the timely delivery of vital safety alerts. [7].
- **Trade-off Between Parameters:** Unsuccessful outcomes, such as an inadequate communication range or an excessive channel load, might emerge from changing a single parameter, such as transmission power or data rate. Simultaneously adjusting several variables improves performance but makes optimisation more difficult.[10]
- **Network Disconnection due to High Mobility:** Multiple link failures and inconsistent connectivity result from vehicles moving at different speeds and using different paths. [8][9].
- **Highly Dynamic Network Topology:** The frequency allocated for VANET communication (e.g., IEEE 802.11p) is restricted and increasing vehicle density leads to fierce competition. [8].
- **Limited Spectrum Resource and Channel Contention:** The network topology is constantly changing because of the great mobility of cars making stable communication challenging [9].

1.4 Proposed Solutions

To overcome the above identified problems or challenges , the integration of Artificial Intelligence (AI), particularly DRL, has emerged as a promising solution [5][6]. DRL, which combines reinforcement learning with deep neural networks, enables agents to make adaptive, real-time decisions based on dynamic environmental conditions without relying on predefined models [6][11]. Various DRL-based techniques have been investigated in recent research to optimise important VANET components. These include dynamic spectrum access strategies to reduce interference and channel contention in dense environments, intelligent trajectory planning for UAV-assisted routing to enhance end-to-end connectivity in sparse networks, and simultaneous adaptation of data rate and transmission power to mitigate channel congestion.

In order to minimise collisions and increase data flow, DRL has also been used for intelligent resource allocation, allowing for dynamic control of channel intervals and backoff techniques. Increasing dependence on DRL to handle routing dependability, scalability, conservation of energy, and spectrum utilisation in VANETs is highlighted by thorough surveys and studies. In a variety of vehicle circumstances, these adaptive algorithms perform better than conventional static or rule-based techniques, providing increased packet delivery ratio, reduced latency, and greater spectrum efficiency. These developments are summarised in this work, which emphasises DRL's contribution to VANET communication method optimisation. The objective is to present an organised summary of DRL applications on VANETs., analyze their effectiveness in addressing core challenges, and identify promising directions for future research.

The overall goal to achieve in this study is:

- To enhance the Packet Delivery Ratio (PDR)
- To keep low latency from start to end
- To maximize network reachability
- To ensure consistent and dependable routing in dynamic and sparse VANET settings [5][6][11].

2. Literature Review

Table 2.1: Literature Survey on Adaptive DRL-Driven Communication in VANET

Sl No	Title	Authors	Year	Methodology	Result	Limitations
1	Adaptive priority and Optimum parameters(AP OP) based relay vehicle selection scheme in VANETs[12]	Virender Kumar, Vivek Srivastava, Nishu Gupta, Jukka Mäkelä	2025	The APOP algorithm optimizes message prioritization and relay vehicle selection for efficient data transmission in VANETs. Performance is evaluated.	Compared to existing methods, the APOP algorithm enhances throughput by 21.7%, reduces delay by 18.3%, and provides 14.6% more data.	High dependence on fixed mobility models limits adaptability; cannot sustain performance under heterogeneous traffic and topology variations.
2	VANET Safety Communications With AHP-Based Resource Allocation [13]	Biraja Prasad Nayak, Lopamudra Hota, Ashok Kumar Turuk, Arun Kumar	2025	AHP prioritizes VUE pairs considering multiple factors; a novel algorithm optimizes resource allocation for V2V communication in VANETs.	Simulation shows ranked VUE pairs outperform reverse-ranked pairs; the approach notably reduces delay for safety-critical messages and improves information value.	Linear resource allocation model struggles with scalability; lacks dynamic adaptation to fast topology changes in urban VANETs.
3	VANETs Routing Optimization: Combining Message Priority, Free Nodes, and Mobility Direction [14]	FANG XU, CHUN-MENG YANG, HAO CHEN, MAN-ZOOR AHMED	2025	(Message Priority, Free Nodes, and Mobility Direction) MPFD is a two-phase routing strategy prioritizing messages and using free nodes for efficient relay selection, evaluated	MPFD strategy boosts delivery speed, reduces delays, and improves ratios.	Message delivery deteriorates in sparse networks due to limited evaluation beyond simulation; lacks real-world vehicular validation.

				through simulations.		
4	Hybrid vehicular access protocol and message prioritization for real-time safety messaging [15]	Mayssa Dardour, Mohamed Mosbah, Toufik Ahmed	2024	Efficient message transmission and deadline guarantees are provided by a hybrid solution that combines the VDA and CSMA/CA protocols with a message prioritization algorithm.	Solution achieved 93% delivery rate with low collision probability.	Evaluation does not include congested intersections; real-time adaptability not proven under mixed traffic compositions.
5	Adaptive Message Prioritization: How to Prioritize VRU Awareness Messages in a Congested V2X Network [16]	Silas Lobo, Andreas Festag, and Christian Facchi	2024	The paper enhances Vulnerable Road Users' safety via prioritized Vehicle-to-Everything communication, integrating VRUs in networks with dedicated awareness messages.	The provided document is the paper's first page, featuring the abstract, so specific quantitative study results are not included.	Clustering method adds communication overhead; feedback latency not addressed; lacks end-to-end network testing.
6	Deep Reinforcement Learning-Based Intelligent Security Forwarding Strategy for VANET [17]	Boya Liu, Guoai Xu, Guosheng Xu et al.	2023	Deep Reinforcement Learning (DRL) with customized state/action space and reward design using Deep Q-Networks (DQN)	Fast convergence, precise adaptive message forwarding selecting secure paths depending on message urgency and reliability; outperforms Q-LBR in delay and security.	Restricted to simplified random-node simulation; lacks model generalization to heterogeneous mobility environments.
7	An improved deep reinforcement	Pratima Upadhyay, Venkat adri Marribona et	2023	Improved Deep Reinforcement Learning (IDRL)	Improved packet delivery ratio by 5-6.5% over peers, lower latency,	Privacy constraints due to GPS dependency; scalability not validated for dense urban topologies.

	learning routing technique for collision-free VANET [17]	al.		leveraging Vehicle-to-Infrastructure (V2I) communication with road clustering transition probability matrix	reduced routing overhead, efficiently adapts to dense/high mobility scenarios.	
8	Deep reinforcement learning techniques for vehicular networks: Recent advances and future trends towards 6G [18]	Abdelkader Mekrahe, Abbas Bradai, Emmanuel Moulay et al.	2022	Comprehensive review and analysis of RL/DRL techniques applied to vehicular networks covering networking, computing, caching, energy, infrastructure management.	Highlights effective DRL algorithms and their uses in managing vehicular resources	Conceptual study; lacks empirical validation; computational cost and training overheads not analyzed quantitatively.
9	Efficient Cache Retrieving Scheme in VANET via Online Reinforcement Learning [19]	Zhenhao Tan, Bin Chen, Ruidong Li, Lei Yang, Xiaohua Xu	2025	A new communication model and algorithm optimize cache scheduling in VANETs, using reinforcement learning to efficiently prioritize data retrieval.	The DRL method optimizes cache scheduling, improving efficiency with fewer waits.	Scalability not tested for multi-zone caching; model assumes stable connectivity unrealistic in real VANET environments.
10	Intelligent Resource Allocation Scheme Using Reinforcement Learning for Efficient Data Transmission	Jin Woo Kim, Jae Wan Kim, Jaeho Lee	2024	Q-learning is a dynamic resource allocation technique that maximises data transmission in VANETs while reducing collisions.	The proposed scheme enhances performance, reduces collisions, and improves packet delivery rates while minimizing overhead in dense networks.	The evaluation overlooks real-world complexities, leading to inefficient traffic management, packet collisions, and inadequate protocols in high-density environments.

	in VANET [20]					
11	DFDRDL: a dynamic fuzzy routing algorithm based on deep reinforcement learning with guaranteed latency. [22]	Yonghong Wang, Marini Othman	2024	DFDRDL optimizes routing in SDNs using DRL, fuzzy logic, traffic prediction, and a deferral mechanism to enhance performance.	DFDRDL excels in latency, throughput, efficiency, and competitive runtime.	Traditional routing struggles with QoS; SDN and DRL face challenges.
12	Improving the performance of Bandwidth Efficient Acknowledgement based Multicast (BEAM) protocol in VANET for Urban environment [22]	Alehegn Minale, Dawit Keflie, and Getamesay Haile.	2024	Through clustering, the suggested approach improves the BEAM protocol by assessing variables including throughput, PDR, and end-to-end delay.	By increasing throughput, PDR, decreasing latency, and permitting V2V communication, the suggested clustering approach improves BEAM protocol performance in VANETs.	Dependence on RSUs restricts emergency responsiveness and causes coverage gaps in RSU-sparse regions.
13	Enhancing Routing Performance Through Trajectory Planning With DRL in UAV-Aided VANETs [23]	JINGXUAN CHEN, DINARUN HUANG, YIJIE WANG	2025	This paper presents a UAV-aided VANET framework enhancing data transmission with adaptive routing, clustering, and optimized UAV trajectory planning.	ADMR protocol improves packet delivery, reduces delay, and excels in scalability.	Consumes extra energy due to UAV operations; scalability sensitive to aerial relay scheduling inefficiency.

14	Challenges and Solutions in VANET Routing: A Study of the Use of Adaptive Algorithms	Reem Jamal Ali	2025	Methodology includes analysis, algorithms, reviews, framework proposal, and metrics.	Adaptive algorithms enhance VANET routing by integrating RL, ACO, and ML, improving delivery ratios, stability, efficiency, and reliability.	Challenges in RL and ML for dynamic VANETs include overhead.
15	AI-ENHANCED ROUTING PROTOCOLS FOR EFFICIENT DATA TRANSMISSION IN WIRELESS NETWORKS	V Anitha, T S Umamaheswari, B Saravanan	2024	The AI-enhanced PARP optimizes routing in wireless networks using DRL, proactively determining paths, evaluated through NS-3 simulations against AODV and DSR.	The AI-enhanced PARP outperformed conventional protocols, improving packet delivery by 23%, reducing delay by 35%, and lowering energy use by 15%.	Traditional routing protocols face performance issues in dynamic networks, while wireless data transmission encounters challenges in reliability and energy efficiency.
16	Systematic review of data exchange for road side unit in a vehicular ad hoc network: coherent taxonomy, prominent features, datasets, metrics, performance	Ahmed Thair Shakir, Md Shabiul Islam, Jit Singh Mandeep	2024	Analysis of RSU trends, strengths, weaknesses, key aspects, and comparison with existing reviews to highlight comprehensive insights and future improvements.	RSUs enhance VANETs by improving communication, safety, and efficiency, while supporting diverse applications and optimizing deployment strategies and performance metrics.	Implementing RSUs faces cost, energy, performance, connectivity, security, and methodological challenges, often lacking comparisons with alternative techniques in studies.

	measures, motivation, opportunities, challenges and methodological aspects					
17	Improving VANET Data Dissemination Efficiency with Deep Neural Networks	Ameur Bennaoui, Mustapha Guezzouri, Mokhtar Keche	2024	Deep learning-based optimized data distribution in VANETs. Detailed vehicle positions, speeds, and traffic densities are available for study in a simulated dataset from OMNET++, SUMO, and OpenStreetMap.	The DNN protocol performed remarkably well in packet delivery, minimized latency, reduced overhead, and demonstrated high precision with low collision rates.	Increased latency due to deep model inference; lacks analysis of performance trade-offs for real-time deployment.
18	A Recent Reinforcement Learning Trend for Vehicular Ad hoc Networks Routing	Alehegn Minale, Dawit Keflie, Getamesay Haile	2023	This paper reviews RL trends in VANET routing, analyzing existing protocols and classifying RL approaches while proposing new methodologies.	This review analyzes relevant RL-based routing protocols in VANETs, highlighting strengths, weaknesses, and the potential of Vehicle-to-Deep Reinforcement Learning.	The review identifies RL-based challenges in VANETs, while the BEAM protocol's limitation is its exclusion of Vehicle-to-Vehicle Communication.
19	A Recent Reinforcement Learning Trend for Vehicular Ad-hoc Networks Routing	Alehegn Minale, Dawit Keflie, Getamesay Haile	2023	This paper reviews RL trends in VANET routing, analyzing existing protocols and classifying RL approaches without proposing new methodologies.	Review examines RL routing protocols in VANETs, noting strengths and weaknesses.	Review highlights RL challenges in VANETs; BEAM excludes Vehicle-to-Vehicle.

20	Inspecting VANET with Various Critical Aspects – A Systematic Review	Syed Irfan Ali Shah, Zhu Jinan, Wang Pengcheng, Muhammad Zafar, Muhammad Arshad	2023	Systematic review identified VANET challenges, solutions, and research gaps.	Comprehensive VANET analysis reveals security, QoS, routing, and research gaps.	Systematic review highlights literature limitations, not own methods.
21	Fuzzy Based Prioritized Message Dissemination and Cooperative MAC Protocol for VANET	M Manoj, A Ch Sudhir	2023	Novel FPMD-CMAC enhances VANET communication through prioritized message dissemination.	FPMD-CMAC protocol excels in message delivery, delay, and throughput.	Computational overhead from fuzzy logic module; limited scalability in dense vehicle conditions..
22	Adaptive Message Prioritization for Vehicular Cooperative Perception at Target Intervals.	MASASHI KUNIBE, REI YA-MAZAKI	2023	Adaptive prioritization method enhances vehicular perception under challenging conditions.	Simulation validates adaptive prioritization for effective vehicular perception.	Non-CAV (Not fully Autonomous Vehicle) determination limits real-world vehicle tracking challenges.
23	A comprehensive survey on data dissemination in Vehicular Ad Hoc Networks	Abdussalam N A, Abdul Jabar H, Nordin R & Hashim	2022	It highlights the efficient Delay-Tolerant ACO for optimal routing and data management.	The method helps improve traffic safety, driving efficiency, and communication.	No universal method exists; challenges remain for efficient, safe, and complete communication.
24	A Performance Analysis of VANETs	Lopamudra Hota, Biraja Prasad Nayak, Arun	2022	Comparison of routing protocols for effective VANET operation.	NS-3 files aid parameter computation and analyze VANET performance	The study's limitations include scope, protocol assumptions, and simplified scenarios.

	Propagation Models and Routing Protocols	Kumar			metrics.	
25	Peer-to-peer overlay techniques for vehicular ad hoc networks: Survey and challenges	Abdelkader Ilyes Ameer, Abderrahmane Lakas	2022	This survey reviews P2P techniques in VANETs, addressing challenges, existing approaches, classifications, detailed descriptions, and comparisons, and future research directions.	Paper examines P2P techniques in VANETs, highlighting strengths and challenges.	The paper highlights VANET challenges: mobility, connectivity, efficiency, security, integration.
26	Adaptive Resource Allocation for Diverse Safety Message Transmissions in Vehicular Networks	Jiayin Chen, Huaguo Wu, Feng Lyu	2022	TARA framework optimizes resource allocation for urban vehicular safety messages.	TARA framework excels in PDR, latency, resource allocation, and adaptability.	TARA framework addresses literature gaps in vehicular resource allocation challenges.
27	Evaluation and improvement of ETSI ITS Contention-Based Forwarding (CBF) warning messages in highway scenarios.	Oscar Amador, Manuel Ureña, Maria Calderon, Ignacio Soto	2022	uses DCC to analyse the ETSI CBF protocol. The following enhancements are suggested: Forward-on-Time (FoT), Geographically-aware Packet Cancellation (GPC), and Duplicate Packet Detection (DPD).	TARA framework excels in PDR, latency, resource allocation, and adaptability.	Limited to highway scenarios. DCC behavior may vary with access technologies. Urban and complex topologies not deeply evaluated.

From the literature survey, the following findings are noted:

- There is a no multi agent communication in previous study and observed that real time learning only applied to simple simulation pattern.
- Lack of unified framework for data transfer mechanism. Huge amount of message transmissions causes network congestion and packet drops, Many of the methods are fail to manage high, low and medium priority messages continuously.
- In large vehicular networks changing grouping techniques and keep on monitoring routing updates struggling from high computational cost
- Legacy Reinforcement Learning techniques struggle to take proper real time decisions.
- In small vehicular network conditions many of the routing decisions fail to handle correct communication routes so which allows maximum packet loss and network discontinuation.
- Older techniques are in proper to handle real time conditions like feedback delay and larger computation cost.

3. Research Gaps

Key Research gaps observed in the Literature Survey:

- *Multi Agent Coordination:* Most of the existing techniques considers vehicle as separate decision taken objects. This lack of coordinated multi-agent intelligence results in redundant message dissemination, inefficient channel utilization. Existing techniques not enable vehicles to collectively optimize forwarding, suppression, and routing decisions based on shared environmental awareness. [23][24][9].
- *Prioritization of Messages:* Existing VANET communication protocols lack a complex and adaptive prioritization mechanism capable of dynamically changing safety critical, traffic related, and non safety messages under live network conditions. Traditional approaches only focus rule based fixed priority assignments that faces the problem of varying networks, channel utilization, and mobility dynamics. [12][13][15][16][31].
- *Bandwidth Efficiency:* Multi transmit based communication and multi route forwarding in VANETs often lead to duplicate packets. While some suppression techniques exist, they are predominantly rule-based and lack contextual intelligence. These mechanisms do not sufficiently integrate message priority, network congestion state, and temporal relevance into duplicate suppression decisions. Causes in effective bandwidth occupancy, maximizing channel utilization and packet loss. Offcourse which degrades QoS to deliver right packet at right destination. [22][8][2][5].
- *Sparse Network Reliability:* When vehicle mobility is high, Observing dynamic topology changes frequently, result in unstable network link disconnections. Diifcult to maintain steady communication in sparse network by the existing routing protocols. The negligence in mobility prediction, route stable estimation, and adaptive recovery mechanisms contributes to high packet loss and increased source to destination delay. [23][24][14][7][9].
- *Delayed Feedback & High Computation:* In traditional routing protocols routing and packet delivery computational cost is high in VANET with the usage of DL and RL techniques because which obsers and takes static decisions. [19][20][21]

4. Objectives

The objectives based on the research gaps are as follows::

1. To develop DRL based adaptive message prioritization by classifying messages into safety-critical, traffic-related, and non-safety categories to ensure timely transmission of urgent information without starving non-critical data.
2. To propose DL based method to improving bandwidth efficiency by detecting and suppressing duplicate messages generated due to broadcast and multi-hop forwarding, thereby reducing redundant transmissions and channel congestion.
3. To create resilient routing methods that maintain reliable communication even when vehicle movement breaks network connectivity in VANETs.

Table 4.1: Mapping of Research Objectives to Research Gaps

Research Objective	Mapped Research Gap(s)	Reason for Mapping
1. To develop DRL based adaptive message prioritization by classifying messages into safety-critical, traffic-related, and non-safety categories to ensure timely transmission of urgent information without starving non-critical data.	• Prioritization of Messages • Multi-Agent Coordination	Ensures timely delivery of critical messages while managing less urgent data. Helps coordinate communication priorities among multiple vehicles.
2.To propose DL based method to improving bandwidth efficiency by detecting and suppressing duplicate messages generated due to broadcast and multi-hop forwarding, thereby reducing redundant transmissions and channel congestion.	• Bandwidth Efficiency	Addresses the issue of unnecessary message duplication to improve network utilization and reduce congestion.
3. To create resilient routing methods that maintain reliable communication even when vehicle movement breaks network connectivity in VANETs.	• Sparse and Dense Network Reliability • Multi-Agent Coordination	Focuses on handling disconnected or sparse and dense network conditions and supports coordinated routing among vehicles for robust data delivery.

5. Methodology

5.1 Overall Methodology

The proposed structure provides adaptive prioritisation, congestion control, and reliable data delivery by using DRL as an intelligent control layer over current communication protocols.

Proposed VANET framework, the decision-making tasks such as message forwarding/suppression, priority assignment, and next-hop selection—involve a discrete action space. DQN is tailored for acquiring optimal policies in environments with discrete actions and provides reduced computational complexity, which makes it ideal for real-time vehicle communication scenarios.

DDPG, PPO, and Actor-Critic are examples of advanced DRL techniques that are better suited for continuous action spaces or have greater training and computing cost. Using these techniques would not yield appreciable extra benefits because your problem does not require continuous control variables. As a result, DQN is chosen as a sensible and effective approach that sufficiently satisfies the goals of scalability and adaptive decision-making in VANETs.

Figure 5.1 depicts the functional flow:

- **Step-1: Input Generation**

- **Vehicle mobility (SUMO)** generates realistic movement of vehicles such as speed, position, direction, and traffic density.
- **Network state (NS-3)** provides communication-related data like packet queues, channel load, collisions, and delays.
- **Message metadata** includes message type (safety, traffic, infotainment), urgency level, timestamp, and sender ID.

- **Step-2: Feature Extraction**

- **Speed & density** reflect traffic conditions
- **Congestion level** indicates network overload
- **Channel busy ratio** measures how occupied the wireless medium is
- **Link quality** estimates reliability of communication links
- **Message urgency/type** distinguishes safety vs non-safety messages.

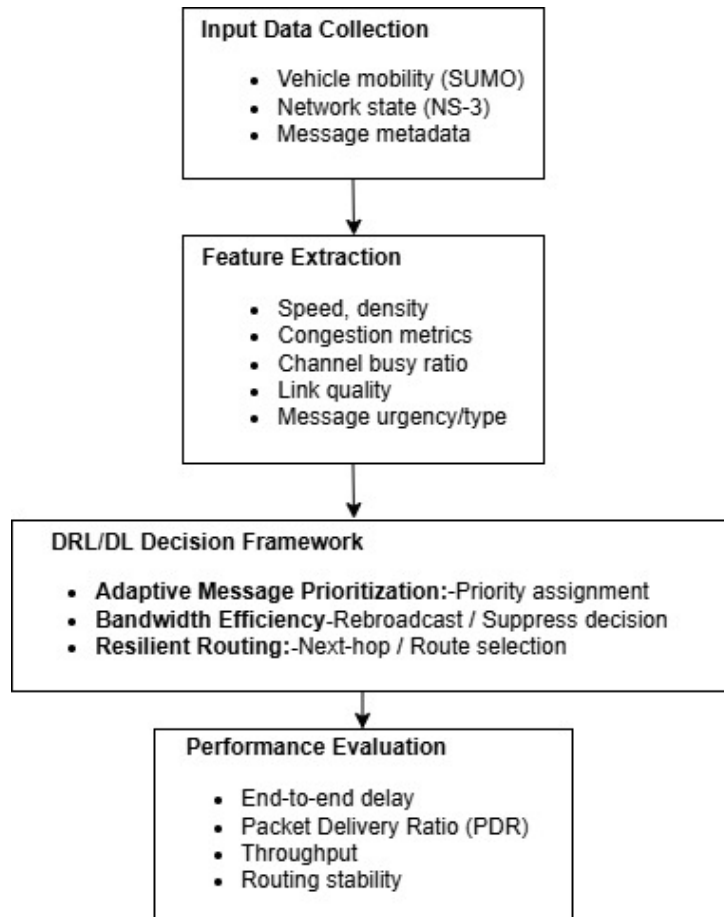


Figure 5.1: Block diagram of the overall methodology

- **Step-3 DRL/DL Decision Framework**

- **Adaptive Message Prioritization:** DRL Learns which messages are critical, assigns higher priority to safety and emergency messages and avoids starvation of non-critical data[16].
- **Bandwidth Efficiency** DL Detects duplicate or redundant packets, Decides whether to rebroadcast or suppress and reduces unnecessary channel usage[1].
- **Resilient Routing** DRL Predicts route stability, selects next-hop based on mobility, link quality and avoids frequent route failures[9].

- **Step-4 Performance Evaluation**

- **End-to-end delay** timeliness of delivery
- **Packet Delivery Ratio (PDR)** reliability.
- **Throughput** network efficiency
- **Routing stability** robustness

5.2 Objective Specific Methodology

The accomplishment of precise, quantifiable goals is the main focus of this section. It entails applying techniques that are intended to directly target and assess advancement towards these particular goals, frequently focused on measurable data and observable results.

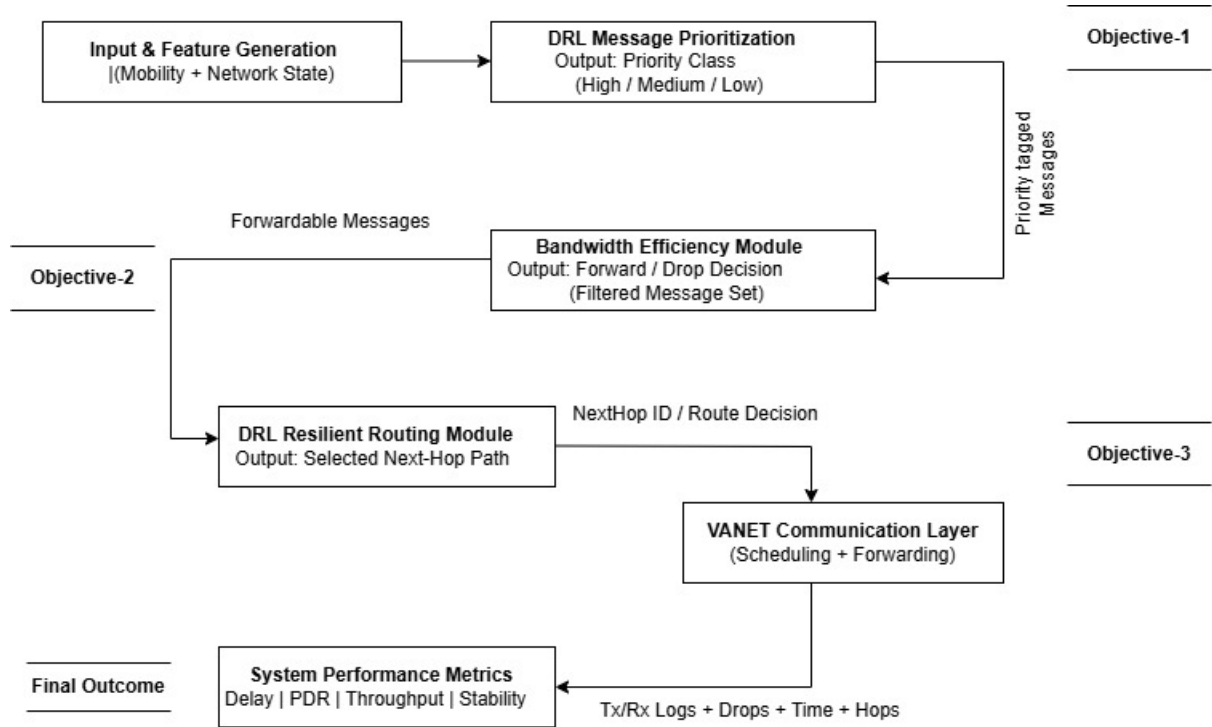


Figure 5.2: Objective wise Integrated Block Diagram

The proposed system first collects vehicle mobility information and current network conditions to generate input features. Then, a DRL-based message prioritization module classifies each message into high, medium, or low priority to ensure urgent messages are handled first (Objective-1). After prioritization, a bandwidth efficiency module filters the messages by deciding whether to forward or drop them in order to reduce congestion and avoid unnecessary transmissions (Objective-2). Next, a DRL resilient routing module selects the best next-hop or route for the forwardable messages by considering link stability and network performance (Objective-3). The VANET communication layer finally schedules and forwards the selected messages through the chosen route. At the end, the system evaluates overall performance using metrics such as delay, packet delivery ratio (PDR), throughput, hop count, drops, and stability, these steps depicted in figure 5.2.

5.2.1 Objective-1:

To develop DRL based adaptive message prioritization by classifying messages into safety-critical, traffic-related, and non-safety categories to ensure timely transmission of urgent information without starving non-critical data..

This goal makes use of DRL techniques like Deep Q-Networks (DQN) and Q-Learning.[17][18].The adaptive message prioritisation process is simulated using SUMO for mobility [37]. It involves making choices frequently and in real time while dealing with dynamically changing network and vehicle situations. A reinforced learning framework is necessary because judgements about message prioritisation have a direct impact on future network conditions (delay, congestion, packet delivery). Therefore, Deep Reinforcement Learning (DRL) is used to learn appropriate priority assignment rules through reward-based feedback and interaction with the environment, which is not possible with traditional Deep Learning (DL) models alone.

Thus, DRL is integrated as the core intelligent decision-making component in the adaptive message prioritization system, enabling dynamic, real-time optimization of communication protocols in the highly mobile and unpredictable environment of VANETs. This use of DRL helps overcome the limitations of traditional static or rule-based protocols by continuously learning optimal policies for efficient message scheduling and prioritization based on evolving network states.

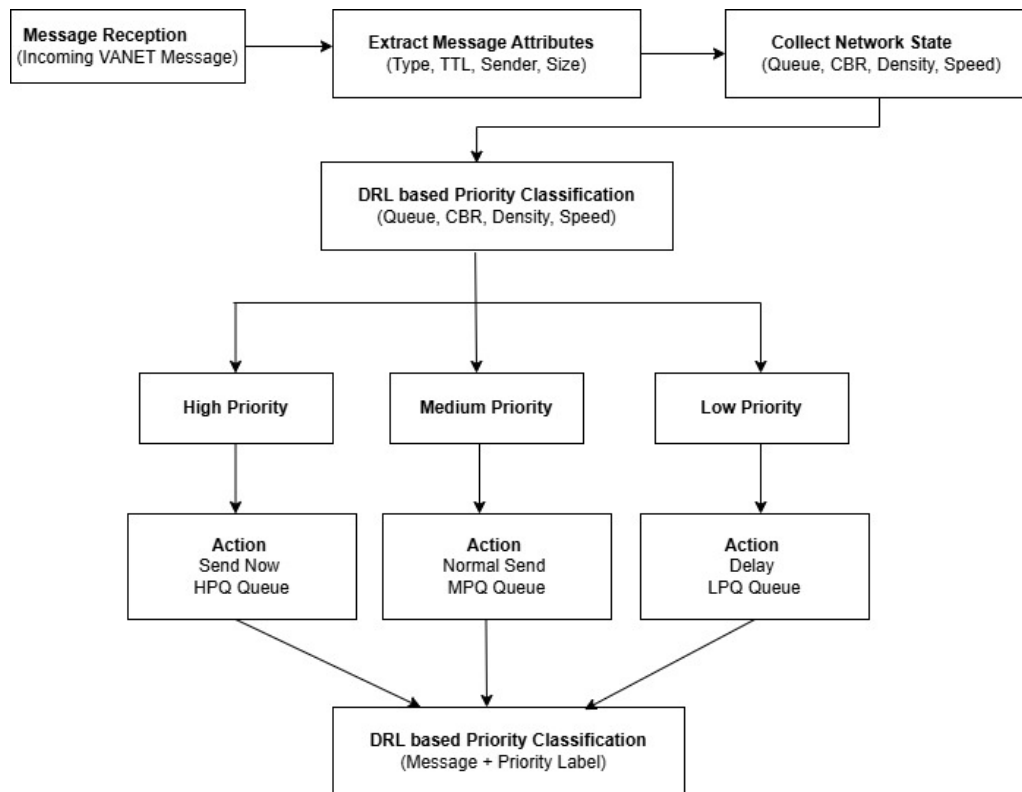


Figure 5.3: DRL based adaptive message prioritization

Figure 5.3 depicts the following functional flow:

- **Step-1: Message Reception**

- An incoming communication from another vehicle is received by a vehicle (or RSU).
- This message can be: accident alert, congestion warning, lane change, infotainment, etc.
- **Example:-** “Accident ahead at 200 meters” from Vehicle-A.
- **Input to system:-** Raw VANET packet/message

- **Step-2: Extract Message Attributes**

- Before deciding priority, the system must know message attributes like: **Type**-Safety / Traffic / Non-safety, **TTL (Time To Live)**-how long message remains valid, **Sender**-vehicle ID / RSU ID, **Size**-packet size impacts transmission time
- **Example:-** For the message “Accident ahead”: Type-Safety critical, TTL-2 seconds, Sender-Vehicle-A, Size-128 bytes

- **Step-3:Collect Network State**

- Because even a high priority message may fail if the network is congested, So the model checks current network conditions.
- Network state parameters: **Queue length (Q)**-messages waiting already, **CBR (Channel Busy Ratio)**-how busy the wireless channel is **Density**-number of vehicles nearby (congestion level), **Speed**-mobility affects link stability
- **Example:-** Queue = 25 messages already waiting, CBR = 0.85 (very busy channel),Density = high (traffic jam area),Speed = 15 km/h (slow moving)

- **Step-4:DRL based Message Priority Classification**

In the proposed framework, incoming messages are classified based on their urgency and application requirements. This classification allows for varied processing at the communication layer to guarantee prompt transmission of safety-critical information

- **High Priority (Safety-Critical Messages):**

- * These messages have direct implications for road safety and must be delivered with minimum latency.
- * **Examples:-**Collision warning alerts, Emergency braking notifications, Accident or hazard alerts, Wrong-way driving alerts, Pedestrian / cyclist detection alerts

- * Message is inserted into HPQ (High Priority Queue), System sends it immediately and it gets channel access earlier than others
- * “Accident ahead” goes to HPQ and is transmitted immediately. **Outcome:**Fast delivery, minimal delay, avoids crash risks

– **Medium Priority (Traffic and Control Messages):**

- * These messages improve traffic efficiency and driving comfort but are not immediately life threatening.
- * **Examples:-**Traffic congestion updates,Road condition alerts (potholes,construction zones),Lane-change assistance messages, Speed limit or signal phase information
- * Message is placed in MPQ (Medium Priority Queue), It is transmitted normally based on scheduling.
- * “Traffic slow near signal”, not urgent like accident, but still useful,goes to MPQ
Outcome:delivered in reasonable time,does not disturb emergency messages

– **Low Priority (Non-Safety Messages):**

- * These messages are delay tolerant and primarily related to user convenience or entertainment.
- * **Examples:-**Infotainment and entertainment content,advertisements,Weather updates, Non-urgent navigation suggestions
- * Message enters LPQ (Low Priority Queue), Transmission is delayed until network is free
- * “Movie streaming / advertisement / music info”, goes to LPQ**Outcome:**prevents channel overload, ensures safety messages are never blocked

The DRL agent observes message attributes + network state and chooses: High Priority / Medium Priority / Low Priority

• **Step-5: Final Outcome**

- The message + its assigned priority label and also the queue assignment:

Treatment of Non Safety Messages

Non safety messages, such as infotainment content, advertisements, weather updates, and comfort-related services, are explicitly assigned the lowest priority level in the proposed communication framework. These messages are delay-tolerant and are transmitted only when sufficient channel resources are available and no safety-critical or traffic-related messages are pending in the transmission queue.

To avoid interfering with the prompt delivery of safety-critical information, non-safety messages are either temporarily suppressed or postponed during times of network congestion or

high vehicle density. By monitoring channel load, queue occupancy, and the existence of higher-priority traffic, the DRL-based prioritisation agent dynamically controls the delivery of non-safety messages. Importantly, non-safety messages are never permitted to take precedence over safety or traffic-control messages, guaranteeing strong enforcement and maintenance of road safety regulations.

By monitoring network circumstances, vehicle density, and message metadata, a DRL-based agent dynamically assigns priority levels to messages, guaranteeing that safety-critical communications are always given priority in channel access and scheduling.

Overall, the methodology's Objective-1 expressly uses DRL techniques as a crucial strategy to improve VANET communication systems' dependability and performance through intelligent adaptive message prioritisation. Main goal of this objective is to automatically classify, prioritise, and schedule messages such that non-urgent (like entertainment or traffic updates) do not delay important data and urgent (like collision warnings) receive instant channel access.

Key tools and techniques used for implement objective-1 is:

- **Simulation:** SUMO (mobility), NS-3 / OMNeT++ / Veins (network)
- **Algorithms:** Q-Learning, Deep Queue Network(DQN)
- **Programming:** Python for ML agent, C++ for simulation
- **Queueing Techniques:** Strict Priority Queue (SPQ), Weighted Fair Queueing (WFQ) [31][32]
- **Protocols:** IEEE 802.11p, Dedicated Short-Range Communications (DSRC) [8].
- **Dataset:** Make use of actual traffic datasets (SUMO mobility traces, for example).
- **Performance Metrics:** PDR, channel utilization, fairness index

Overall objective-1 methodology technique solves the fundamental problem of simultaneous transmission of mixed-priority data in dynamic VANET environments by guaranteeing that safety-critical messages are always delivered within stringent latency boundaries, enhancing network efficiency, and avoiding bandwidth wastage.

5.2.2 Objective-2:

To propose DL based method to improving bandwidth efficiency by detecting and suppressing duplicate messages generated due to broadcast and multi-hop forwarding, thereby reducing redundant transmissions and channel congestion.

Main goal is to reduce the unnecessary network traffic carried on by redundant or repeated message distribution, which is a significant cause of congestion in dense VANET environments. Rather than using rule-based or reinforcement learning techniques, a Deep Learning-based

solution is used because duplicate detection is really a pattern recognition and classification problem.

Sequential decision-making or reward-based policy optimisation are not necessary for the suppression of duplicate and redundant messages. Rather, it entails identifying patterns in communication data from the past, including distribution behaviour, temporal similarity, and message repetition frequency. As a result, Deep Learning (DL) is better suited for this goal than Deep Reinforcement Learning (DRL) since it allows for faster inference and accurate message classification with less computing complexity.

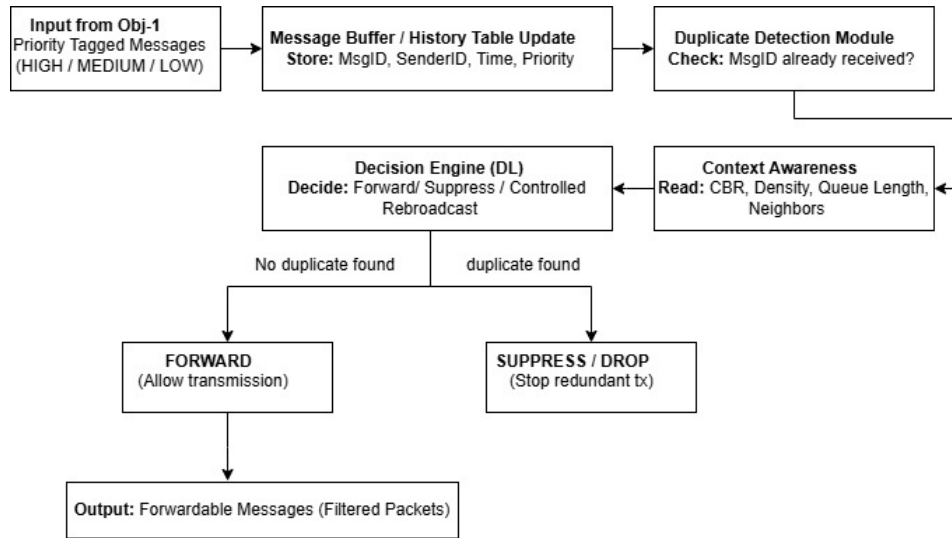


Figure 5.4: Workflow for DL based Bandwidth Efficient Communication Model

Figure 5.4 depicts the following functional flow:

- **Step-1 Input from Obj-1**

- Every incoming message is given a priority level (High, Medium, or Low) determined by its importance and network situation. These messages marked with priority act as the direct input for Objective-2
- Each received message contains: Message packet (payload + header), Priority tag (HIGH / MEDIUM / LOW), Message identifier (MsgID), Sender information and timestamp
- **Output:** An updated message history table containing recent message records.

- **Step-2 Message Buffer / History Table Maintenance**

- Each vehicle maintains a local message history buffer to keep track of recently received messages. This buffer functions as a short-term memory for detecting repeated transmissions.

- The buffer stores the attributes like: Message ID (MsgID), Sender ID, Arrival timestamp, Message priority
- This buffer allows the system to determine whether an incoming message is: **New (fresh)** or **Duplicate (already received earlier)**, By maintaining this table, the system avoids unnecessary rebroadcasting of identical information.
- **Output:** An updated message history table containing recent message records.

- **Step-3 Duplicate Detection Module**

- The duplicate detection module compares the current message against entries in the message history buffer.
- A message is marked as duplicate if same MsgID already exists in the buffer or same sender reports the same event within a predefined time window.
- If neither condition is satisfied, the message is considered fresh.
- **Output:** A binary duplicate indicator:

$$\text{Duplicate_Flag} \in \{\text{YES}, \text{NO}\}$$

- **Step-4 Context Awareness and Network State Evaluation**

- Duplicate status alone is insufficient for making forwarding decisions. Hence, the system evaluates the current network context to support intelligent decision-making.
- This step observes the parameters like Channel Busy Ratio (CBR), Neighbor density, Queue length / buffer occupancy, Overall congestion level.
- **Output:** A set of network context features provided to the decision engine.

- **Step-5 Decision Engine(DL)**

- It integrates: Priority information (from Objective-1), Duplicate detection result, Network context parameters
- Based on this combined input, the system selects one of the following actions:
 - * **Case A: Forward:** A message is forwarded if: -It is fresh (not a duplicate) or It is a duplicate but marked as HIGH priority, where reliability is more important than bandwidth saving.

Action The message is allowed to proceed to Objective-3 for routing

- * **Case B: Suppress/Drop:** A message is suppressed if:-It is a duplicate, Priority is MEDIUM or LOW, and The channel is congested or buffers are overloaded

Action The message is discarded, preventing unnecessary rebroadcast.

- **Output:**A final forwarding decision:

$$\text{Decision} \in \{\text{Forward}, \text{Suppress}\}$$

• Step-6 Action Execution

- Based on the decision:
 - * **If Forward:** The packet is placed in the forwarding queue, Marked as forwardable and Passed to Objective-3 for routing
 - * **If Suppress:** Rebroadcast is stopped and Packet is discarded
- **Output:**Unnecessary transmissions are eliminated before routing, saving bandwidth and reducing congestion.

• Step-7 Results

- The final output of Objective-2 is a filtered set of forwardable messages. These messages: Are non-redundant or justified duplicates, Are priority-aware and Are suitable for routing and forwarding
- **Output:**Forwardable Messages → Input to Objective-3 (Resilient Routing)

Key tools and techniques used for implement objective-2 is:

- **Simulation:** SUMO (mobility), NS-3 / OMNeT++, MATLAB
- **Learning Model:** Deep Neural Network (DNN), Supervised learning approach
- **Programming:**Python for DL development and C++ for simulation
- **Baseline Methods for Comparison::** Hash-based duplicate detection, Bloom filter-based suppression
- **Performance Metrics:**Throughput, Packet Delivery Ratio (PDR), Bandwidth Utilization, Duplicate Suppression Ratio.[8][9]

5.2.3 Objective-3:

To create resilient routing methods that maintain reliable communication even when vehicle movement breaks network connectivity in VANETs.

This mainly aims to solve the problem of network breaks that happen when vehicles move fast and change positions in a VANET. When connections are lost, the proposed routing method helps continue data transmission by storing data temporarily (buffer-carry-forward), finding new routes (route recovery), and predicting vehicle movement (mobility prediction).

Simulation results show that this routing system works well in both heavy traffic and low traffic conditions. It improves important performance measures such as faster data delivery (lower delay), higher successful packet delivery (PDR), and quicker recovery when the network gets disconnected.

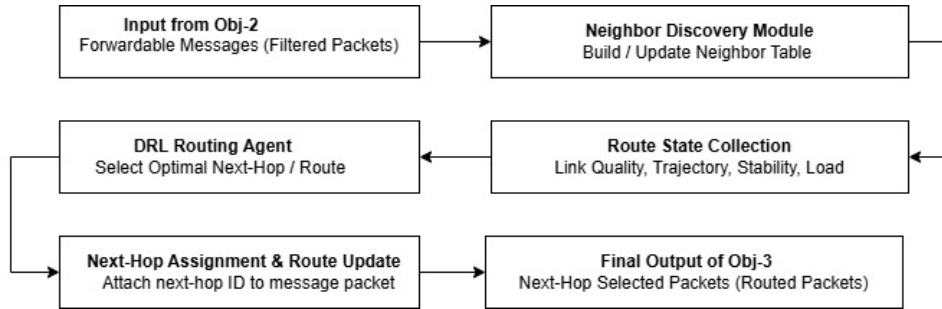


Figure 5.5: DRL-Based Resilient Routing and Next-Hop Selection

Figure 5.5 shows working flow DRL-Based Resilient Routing and Next-Hop Selection with following steps

– Step 1: Input from Objective-2 – Forwardable Messages:

- * After Objective-2, only valid and non-redundant messages remain.
- * It Receives forwardable / filtered packets, These packets are already priority aware (from Obj-1) and redundancy controlled (from Obj-2)
- * This step is important because routing decisions are made only for messages that are allowed to be transmitted, avoiding unnecessary route computation.
- * **Output:** Forwardable messages ready for routing

– Step-2 Neighbor Discovery Module:

- * Each vehicle identifies nearby vehicles within communication range.
- * Periodic beacon messages are exchanged, A neighbor table is built or updated

- * Information stored in neighbor table like Neighbor vehicle ID, Distance & position, Speed & direction and Signal strength / link quality
- * This step is important because routing in VANET is hop-by-hop, so knowing current neighbors is essential for selecting a reliable next-hop.
- * **Output:** Updated Neighbor Table

– **Step 3: Route State Collection:**

- * This step prepares the routing state needed by the DRL agent.
- * Parameters collected: Link quality / signal strength, Estimated link lifetime, Vehicle trajectory (speed + direction), Local traffic load / congestion and Hop count estimate
- * This step is important because VANET links are highly dynamic. Using these parameters helps predict route stability.
- * **Output:** Routing state vector (environment state)

– **Step 4: DRL Routing Agent – Next-Hop Selection:**

- * This step takes the routing state as input, evaluates available neighbors
- * Selects the best next-hop that maximizes link stability, packet delivery success and low route breakage
- * It choose one neighbor from the neighbor table as next-hop
- * This step is important because traditional routing fails under high mobility and DRL learns from experience and adapts to changing traffic and topology.
- * **Output:** Selected next-hop vehicle ID

– **Step 5: Next-Hop Assignment & Route Update**

- * Once the next-hop is selected, the routing decision is applied.
- * Next-hop ID is attached to the message packet, hop count and routing header are updated and finally packet is marked as “ready for forwarding
- * This step is important because it converts the routing decision into an actionable packet, enabling actual transmission.
- * **Output:** Routed packet with next-hop information

– **Step-6 Results**

- * The final output of Objective-3 is Next hop selected packets, Also called routed packets

- * These packets are now sent to VANET Transmission Layer for actual forwarding.
- * **Output:** Next-Hop Selected Packets

6.Expected Outcome

The expected outcome is a complete, evaluated and verified framework that enables reliable and bandwidth-efficient communication for VANETs regardless of difficult circumstances like high vehicle mobility, frequently occurring disconnections, and differing traffic densities..

Expected results are outlined below:

- **Robust Message Prioritization and Scheduling:**Critical safety alerts have priority in a predictive message classification.
- **Dependable Data Transport in Disjoint Networks:**An improved mixed protocol for routing that employs movement estimation and store-carry-forward methods may produce accurate data routes for delivery even in conditions with minimal density or high mobility, where typical routing strategies would fail. This ensures that even in the case of a network outage, information will continue to flow.
- **Bandwidth Efficiency through Redundant Data Elimination:**A efficient module to identify and blocking duplicate messages will significantly cut down on duplicate transmissions over a network. decreased channel utilization, less data packet collisions and a more effective utilization of the VANET's restricted bandwidth are the immediate effects of this enhancement.
- **Integrated Cross-Layer Framework:**Instead of working separately, the network stages work together as one whole. The application layer decides if a normal or emergency message is being sent. For the purpose of to prevent collisions with data, the MAC layer controls when vehicles can transmit. Strong routes are identified and broken connections are quickly fixed through the network layer.

7.Importance of Proposed Research

7.1 Significance of Proposed Research

The proposed research study proposes a comprehensive communication framework for VANETs that is robust, bandwidth-efficient, and prioritization-aware. Its ability to manage several essential issues simultaneously and ensure the dependable delivery of safety-critical information is what makes it so important, particularly in scenarios where networks are fragmented and evolving rapidly.

7.2 Key Contributions

1. **Enhanced Road Safety:** The main aim is to contributing the main objective of ITS by ensuring guarantee delivery of safety alerts. Of course, which prevents accidents, reduces traffic incidents, and saves lives.
2. **Improved Reliability in Fragmented Networks:** Due to frequent link interruptions on vehicles,speed movement, packet drops and unstable communication are common in VANETs. The proposed system integrates mobility guesing, mixed path selection, and store and forward methods to ensures reliable data transfer even in the case of disconnected connectivity.
3. **Bandwidth Efficiency via Duplicate Suppression:** Duplicate messages detected and takes proper actions on those duplicate messages based on the priority levels, whether it has been discarded or forwarded.

8. Research Plan Schedule

8.1 Overall Schedule

Research Activity	Time in Months (from Registration)								
	06	12	18	24	30	36	42	48	54
Course Work									
Literature Survey									
Objective 1									
Objective 2									
Objective 3									
DAC presentation and registration renewal									
Conference presentations									
Journal Publications									
Preparation of Final Thesis									
Synopsis presentation									
Submission of Thesis									

Figure 8.1: Research Plan Schedule

9. Pilot Study

Pilot study aimed to validate the feasibility of using DRL for message prioritization in VANETs. Specifically, the study tested whether a DRL agent can learn to distinguish between SAFETY (urgent) and NON-SAFETY (non-urgent) messages in a simulated mobility environment, and assign appropriate transmission priorities under bandwidth-constrained conditions.

9.1 Obj-1 Pilot Study: Adaptive message prioritization

– Methodology

* Synthetic Data Generation

- **Vehicular mobility traces** were generated using SUMO, producing vehicle positions, speeds, and movement patterns.
- **A Synthetic message dataset** was developed, comprising NON-SAFETY messages (such as infotainment and marketing) and SAFETY messages (such as collision warnings and brake alerts).
- **A Synthetic message dataset** was developed, comprising NON-SAFETY (such as infotainment and marketing) and SAFETY (such as collision warnings and braking alerts) messages.
- **Messages** were aligned with vehicle mobility traces to form an integrated dataset for DRL training.

* Data Pre-processing

- Both datasets were merged into a unified format and ensuring that each timestep contained vehicle mobility features $\mathbf{m} = (x, y, v)$ and message metadata $\mathbf{d} = (\text{msg_type}, \text{subtype})$.
- SAFETY vs NON-SAFETY classes were explicitly labeled for training.

* DRL Environment Design

- A specific The gymnasium setting had been set up, in which the agent's state consisted of vehicle movement variables and communication type.

- Actions represented **priority levels** (Low, Medium, High).
- Reward function helps agent to assign **High priority** to SAFETY messages, and avoid unnecessary prioritization of Non safety traffic.

* **Training and Evaluation**

- A **DQN** agent was trained on the merged dataset.
- **Classification accuracy, average reward, and accuracy by message type (SAFETY vs. NON-SAFETY)** were among the evaluation metrics.
- For analysis, performance plots (**reward trends, action distributions, accuracy charts**) were created.
- **Summary Metrics:** Overall Accuracy is 98.54%, Non-Urgent Accuracy is 98.38%, and Urgent Accuracy is 100%.

– **Key Findings**

- * After adjusting the message generator balance, the DRL agent successfully **learned to prioritise messages**, with high overall accuracy ((90%)).
- * The pipeline produced comprehensible results from mobility simulation to DRL training.
- * Class imbalance problems (no SAFETY messages) were discovered during early runs, however these were fixed by parameterising the message generator.
- * The DRL agent maintained efficiency for non-urgent communications while demonstrating **significant improvement in urgent message accuracy** with corrected balance.

- **Results** The DRL agent successfully distinguished between safety-critical and non-safety messages with high accuracy, ensuring that urgent alerts consistently received higher transmission priority.

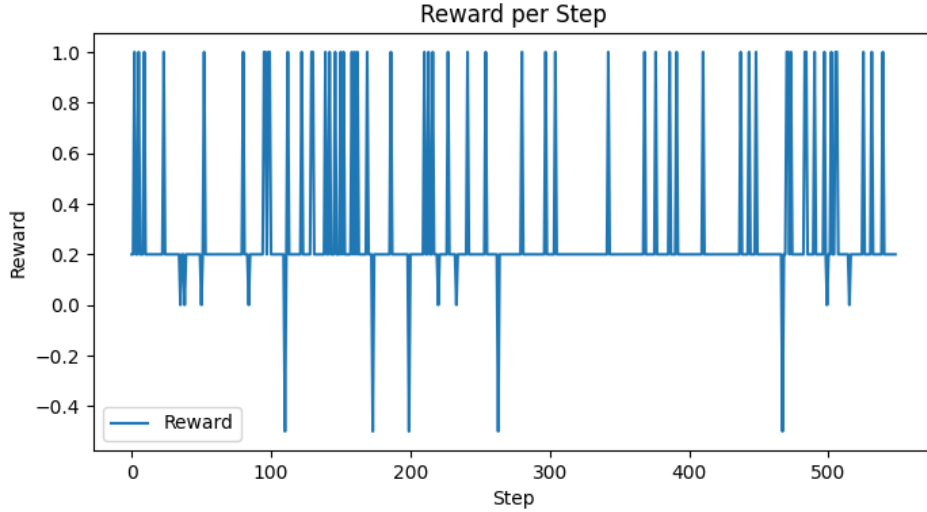


Figure 9.1: Reward trend over training episodes for the proposed model

Figure 9.1 shows the learning behaviour of the DRL agent during adaptive message prioritization. Initially, rewards fluctuate due to random decisions. Over time, rewards increase and stabilize, indicating that the agent successfully learns optimal priority assignment for safety critical messages under dynamic VANET conditions. Each **step** represents one learning interaction where the DRL agent observes network conditions, assigns message priority, and receives a reward based on correctness. The increasing **reward** values indicate that the DRL system gradually improves from poor decision-making to consistently correct message prioritization, proving successful learning.

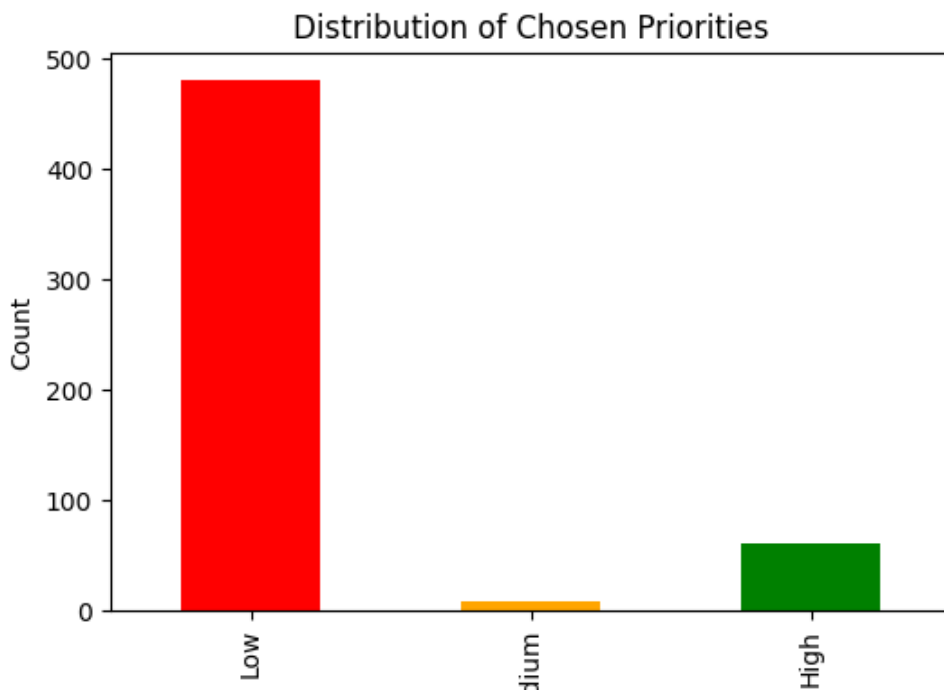


Figure 9.2: Distribution of actions taken by the DRL agent during training

Figure 9.2 demonstrates that the DRL agent learned an intelligent priority policy, where safety critical messages most of the time assigned with high priority, traffic related messages received moderate priority, and non safety messages were kept at low priority.

9.2 Obj-2 Pilot Study: Bandwidth-efficient Communication Protocol

– Methodology

* Data Pre-processing

- Dataset contains message and vehicle movement data and even messages are identified with the label new or duplicate.

* Duplicate Suppression Mechanism

- Duplicates are detected based on the unique vehicle ID, packet delivery time and type of message received.

* Evaluation Metrics

Duplicate suppression performance mechanism were measures with the metrics like:

- **Throughput:** The number of messages successfully forwarded per unit time.
- **PDR:** The ratio of forwarded messages to the total number of generated messages..
- **Duplicate Suppression Ratio:** The proportion of duplicate messages that were successfully identified and dropped relative to the total number of messages received.

– Experiment Scenarios

- * Two types of experimental conditions were examined. The first varied vehicle density between 20 and 100 vehicles to represent sparse and dense networks. The second varied duplicate message probability at 10% and 50% to study different redundancy levels. These scenarios enabled analysis of how network size and duplication intensity affect suppression performance.

– Results and Discussion

- * The results demonstrated that the suppression mechanism performed as expected. When vehicle density and duplication were low, high throughput and

packet delivery were achieved with few dropped messages. As density and duplication increased, network load reduced throughput and delivery performance, but the system effectively removed more redundant messages, improving suppression efficiency.

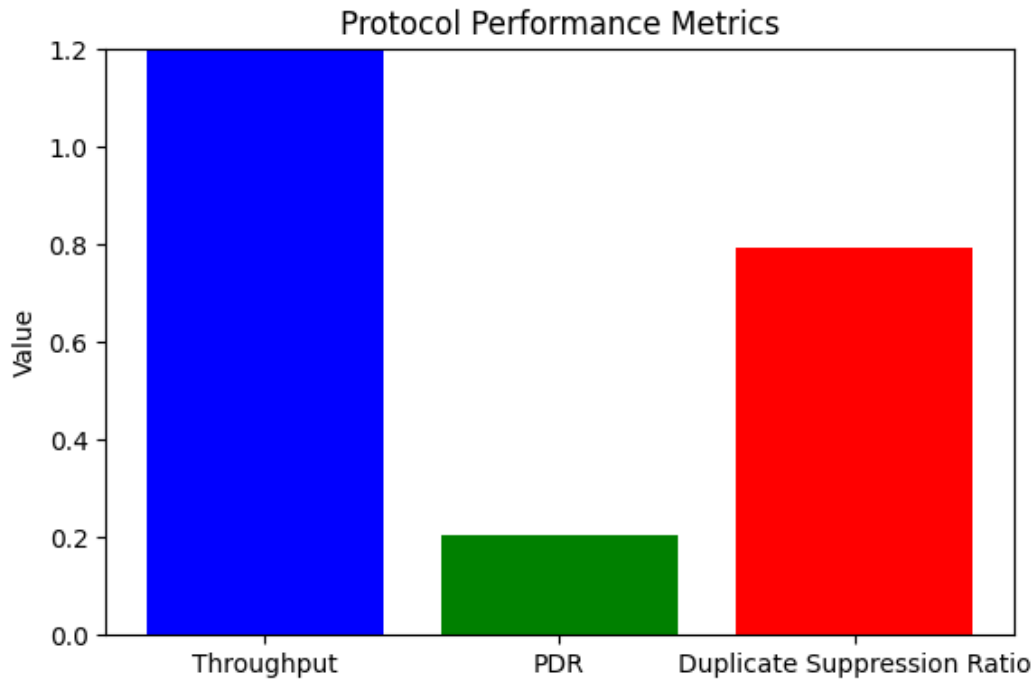


Figure 9.3: Evaluation of protocol performance metrics

Figure 9.3 shows that as vehicle density and duplicate messages increase, throughput and packet delivery ratio gradually decrease due to higher congestion. However, the proposed duplicate suppression mechanism effectively removes redundant traffic, preventing severe performance degradation and maintaining stable network efficiency, especially in highly congested scenarios.

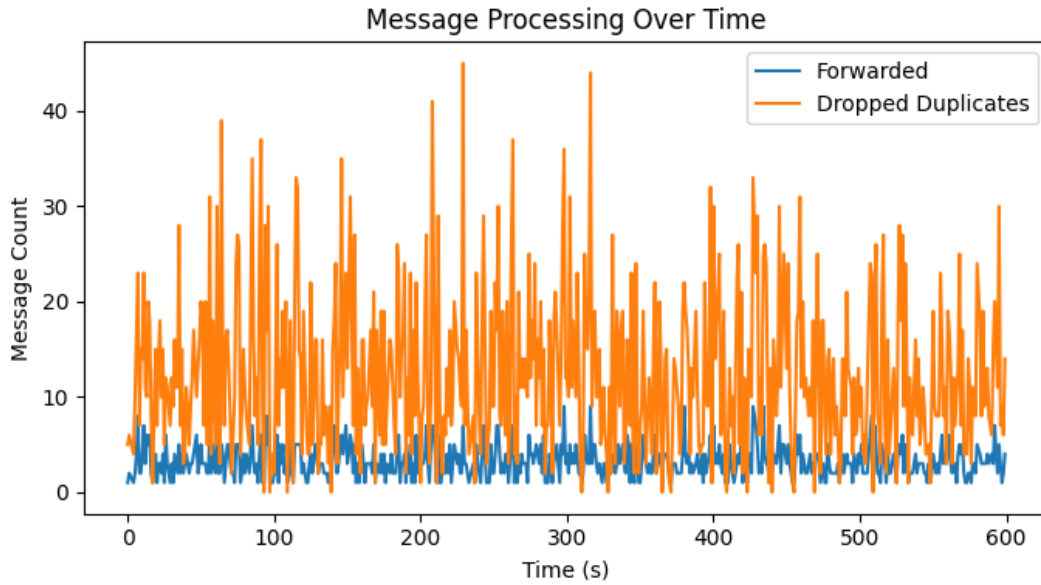


Figure 9.4: Time-series analysis of message processing: forwarded messages vs. dropped duplicates

Figure 9.4 presents a time-series comparison between forwarded messages and suppressed duplicate packets during simulation. Initially, most incoming messages are forwarded due to low redundancy levels. As traffic intensity increases over time, the number of duplicate messages rises; however, the proposed suppression mechanism effectively identifies and drops redundant transmissions.

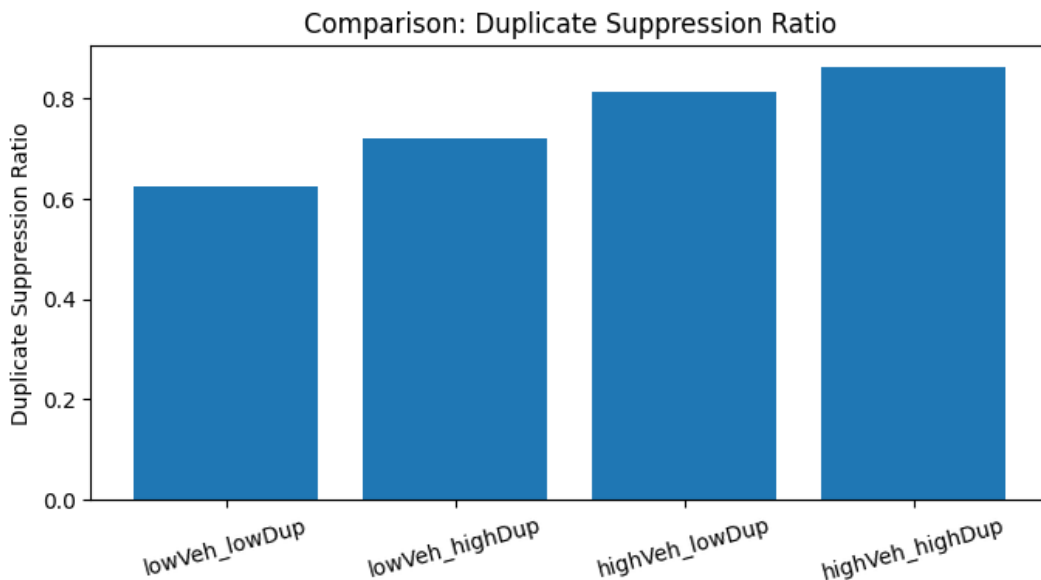


Figure 9.5: Impact of vehicle density and duplicate traffic on Duplicate Suppression Ratio

Figure 9.5 demonstrates a direct relationship between vehicle density and duplicate suppression ratio. In small density road, redundancy is less, hence low duplicates occurs. In

large density, repeated transmissions become more frequent, leading to more duplicate messages. In highly dense networks, the suppression mechanism removes a substantial proportion of redundant messages, significantly reducing congestion.

9.3 Obj-3 Resilient Routing and Data Delivery

- **Methodology**

- The main research aim is to observe the reliable data transmission in VANET.
 - * To solve the major problems in previous study and observations, the proposed approach first **predicts link quality** using vehicle location and movement speed information, allowing the system to observe which is the active links and disconnected one. Based on this prediction, the routing mechanism selects best stable paths apart from poor path.
 - * When the network becomes temporarily disconnected due to vehicle movement (a situation known as fragmentation), the system does not immediately drop messages. Instead, it uses a **buffer-carry-forward strategy**, where vehicles store undelivered messages in memory and physically carry them until a new communication link becomes available.
 - * Once connectivity is restored, the stored messages are **forwarded to the next vehicle or destination**, and if any packets were lost during the disruption, **retransmission and recovery mechanisms** help ensure successful delivery.

- **Experimental Scenario**

- **Simulation Environment:** SUMO for vehicular mobility traces, NS-3 for network communication.
- **Simulation Area:** Urban grid topology (2 km × 2 km).
- **Vehicle Density:** 50, 75, and 100 vehicles.
- **Mobility Speed:** 20–60 km/h.
- **Traffic Type:** Mix of safety-critical and non-safety messages.
- **Evaluation Metrics:**
 - * Packet Delivery Ratio (PDR)
 - * End-to-End Delay (ms)

* Routing Overhead (%)

* Fragment Recovery Rate (%)

– Experimental Scenario

* **Graphical Results:**

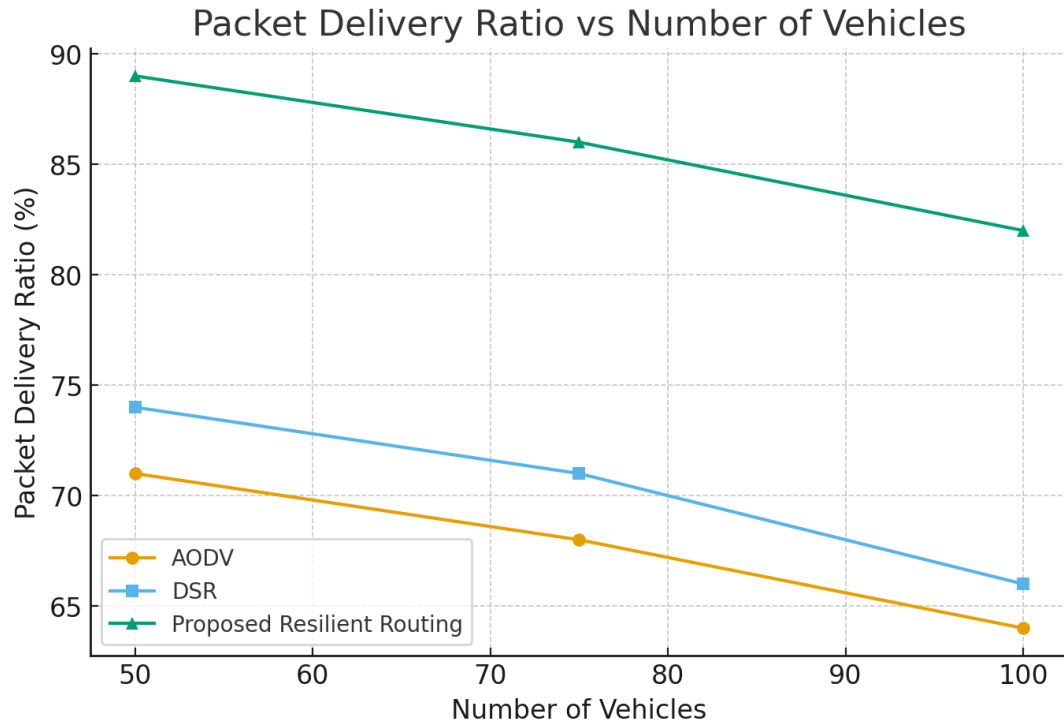


Figure 9.6: Relationship between the number of vehicles and the packet delivery ratio.

Figure 9.6 shows that as the number of vehicles increases, packet delivery improves for all protocols, but the proposed resilient routing consistently achieves higher delivery than AODV(Ad hoc On Demand Distance Vector) and DSR(Dynamic Source Routing), especially in sparse and fragmented networks where traditional protocols suffer from frequent route failures.

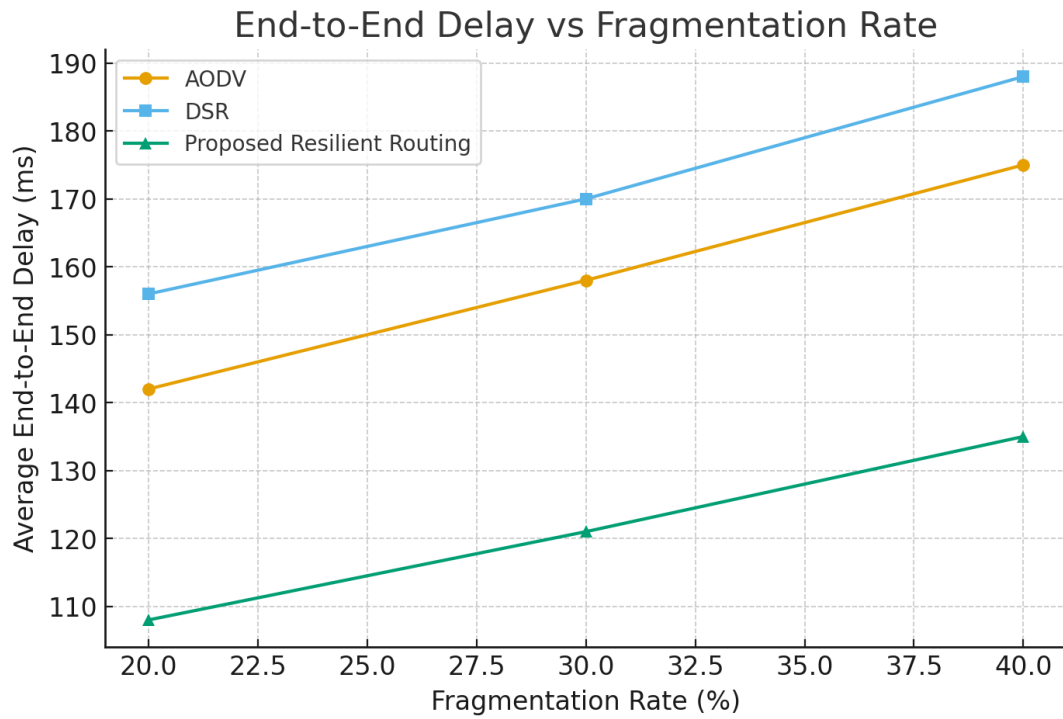


Figure 9.7: End-to-End Delay vs Fragmentation Rate

Figure 9.7 illustrates the effect of network fragmentation on end-to-end delay. As fragmentation rate increases, message delivery time rises due to frequent route breakages and temporary disconnections. The observed trend confirms that fragmented VANET environments significantly impact communication latency, highlighting the importance of resilient routing mechanisms such as buffer-carry-forward and route recovery to maintain reliable data delivery.

10.Course Work Details

Table 10.1: Course Work Details suggested by DAC with credits

SI No	Course Name	Credits	Status
1	Research and Publication Ethics	2	Completed
2	Advanced Research Methodology (ARM)	4	Ongoing
3	Vehicular Adhoc Networks (VANET)	3	Yet to start
4	Deep Reinforcement Learning (DRL)	3	Yet to start

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